

Eric Gitangu

Software Engineer, Payments Platforms and Distributed Systems

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PROFESSIONAL SUMMARY

Software engineer with 11 years in production systems, the last five in African fintech, mobility and energy access. I currently own platform architecture and data security for a mobility and payments company: five web applications and a mobile API on one backend, 6,800 unit tests, 280 architecture decision records, an atomic release process with automatic rollback, and card and mobile money integrations (Mastercard Payment Gateway Services, M-Pesa, MTN, Tigo) built within PCI DSS SAQ-A scope. Before that I ran CI/CD and test automation for a pay-as-you-go solar platform in seven countries, founded an ML security company for medical device software, and led a digital addressing pilot with a county government. I work in TypeScript, Java, PHP, Python, Rust, Go and C#. I design event-driven systems, plan for failure, and instrument what I ship. I set technical direction for the team, review code daily, and have mentored 15 engineers.

EXPERIENCE

Senior Full Stack Engineer, Platform Architecture and Data Security, Moovn Technologies

Feb 2026 – Present

Nairobi, Kenya (Seattle HQ)

Platform architecture and data security for a mobility and payments company operating in East and Southern Africa.

- Own the architecture of five web applications and a mobile API served from one PHP backend. The codebase carries 6,800 unit tests and 280 architecture decision records.
- Designed the release process. Each release is an immutable directory activated by a symlink flip. A post-flip smoke test rolls back automatically on a real error. Deploy is one command and finishes in under a minute; rollback is one command.
- Built the card top-up rail on Mastercard Payment Gateway Services hosted sessions, so card data never reaches our servers, keeping the platform in PCI DSS SAQ-A scope. Built M-Pesa STK and B2C, MTN and Tigo mobile money rails with per-country routing and idempotency that fails closed.
- Designed a trust score engine that ingests signed, idempotent events from partner products into a hash-chained ledger and recomputes scores nightly for 40,000 users. Rolled it out behind feature flags to the admin console, rider portal and mobile API.
- Hardened access and change control: key-only SSH with fail2ban, WebAuthn and TOTP MFA for admins, secret scanning and test gates in git hooks, pentest remediation, and encrypted nightly off-site database backups with restore drills.
- Added Prometheus, Grafana and Loki to a legacy application without changing application code. Traced a database failover storm to a mis-sequenced reboot and fixed the liveness checks. Wrote the team's release and deployment guides.

PHP TypeScript Next.js React Node.js MongoDB Redis Nginx Prometheus Grafana

Team Lead, QA and Engineering, Ignite Energy Access (formerly ENGIE Energy Access)

May 2025 – Jan 2026

Nairobi, Kenya

CI/CD and test automation for a pay-as-you-go solar platform in seven countries.

- Owned CI/CD pipelines and test infrastructure for Kenya, Uganda, Tanzania, Rwanda, Mozambique, Côte d'Ivoire and Zambia. Defined the testing standards and release gates engineering adopted.
- Built end-to-end test suites in Playwright and Cypress for payment processing, token generation, subscriptions and IoT device provisioning. Regression moved from a multi-day manual cycle to an automated run on every pull request.
- Led the Rails 5 to 7.2 upgrade across the payment, PAYG token, IoT diagnostics, subscription and firmware update systems, including multi-tenant configuration for each market.
- Designed asynchronous token generation on RabbitMQ and a Redshift pipeline for payment events, which removed timeouts during peak billing. Managed a team of QA engineers.

Ruby on Rails Playwright Cypress RabbitMQ Redshift PostgreSQL GitLab CI Docker Kubernetes K6

Director of Engineering / Lead Engineer, Vishnu Systems, Inc.

Jan 2024 – Aug 2025

Atlanta, GA · Remote

Founded the company and built its product: ML security testing for software in medical devices. Ran concurrently with Baw Bab.

- Built the platform from scratch. It uses CNNs, retrieval-augmented generation and graph neural networks to detect adversarial attacks on ML models and to report gaps against HIPAA, FDA 21 CFR Part 11 and IEC 62304.
- Designed the AWS backend in Python and Rust (PyO3) with a Go middleware layer for routing, rate limiting and fan-out to evaluation workers. Circuit breakers and back-off kept a live degradation invisible to customers.
- Ran ML pipelines on Airflow with MLflow for experiment tracking, model registry and reproducibility.

Python FastAPI Rust PyO3 Go AWS Kafka Airflow MLflow PyTorch

Lead Software Engineer, Baw Bab Technologies, Inc.

Apr 2023 – Aug 2025

Seattle, WA / Nairobi, Kenya

Digital addressing for last-mile delivery in informal urban areas. Ran concurrently with Vishnu Systems.

- Built computer vision pipelines over Azure, Google and OpenStreetMap satellite imagery to detect residential structures and assign them Google Plus Codes.
- Ran a pilot with the Nairobi County Government, the second digital addressing programme in Africa, integrated with county land registry records.
- Owned the stack: React, Node.js, Kafka and MongoDB, with OpenAPI contracts and a Go backend-for-frontend separating the field agent mobile app from the county web console. Hosted on Azure, later Azure and GCP Cloud Run.

React Node.js Go Kafka MongoDB PostGIS Azure GCP Cloud Run OpenAPI

Senior Software Engineer, Reinsurance Group of America (RGA)

Jan 2022 – Jul 2022

Remote

- Worked on the Java microservices platform for actuarial, billing and reinsurance treaty processing. Defined service boundaries and data contracts with the architecture team, wrote integration tests and PostgreSQL migrations, and introduced OpenAPI and contract tests so services could deploy independently.

Java Spring Boot PostgreSQL OpenAPI Pact

Independent Engineer & Mentor, Deveric Technologies

Jan 2019 – Feb 2022

Nairobi, Kenya / Remote

- Founded a mentoring programme: 15 developers mentored through code review, system design sessions and career coaching, 6 of whom moved to mid or senior roles. Completed 80 technical certifications in cloud, backend, blockchain, AI and security. Built Go, Rust and Python projects for hackathons and open source.

Go Rust Python TypeScript AWS GCP Azure

Software Engineer, Vecima Networks Inc. (now Concurrent)

Jan 2018 – Jan 2019

Peachtree City, GA

- Wrote C and embedded assembly for petabyte-scale video streaming pipelines with deterministic store-and-replay and zero frame loss. Designed redundant data paths and self-healing recovery for content delivery.

C Assembly Linux CDN

Software Engineer, Veracode

Dec 2015 – Jan 2018

Burlington, MA

Application security platform. Integrated static, dynamic and composition scanning into customer CI/CD pipelines.

- Built the Jenkins, GitLab CI and Azure DevOps integrations that let customers run security scans inside their build pipelines. Worked on the monolith-to-microservices decomposition on Spring Boot, JPA and Hibernate with HATEOAS REST APIs.
- Owned the Visual Studio plugin (.NET on Windows), including the TLS 1.2 enforcement upgrade for customers in regulated industries. Reviewed every feature against the OWASP Top 10 and NIST SSDF.

Java Spring Boot JPA/Hibernate C#.NET REST Jenkins OWASP

SELECTED PROJECTS

PesaCore, a .NET 10 core banking platform · [pesacore-web-261341881894.africa-south1.run.app](#)

Core banking platform on ASP.NET Core 10 with a Blazor front end. CQRS with idempotent transfers, PostgreSQL and Redis, OpenTelemetry into Prometheus, Grafana and Jaeger, a hardened nginx edge. Runs on GCP Cloud Run with 67 tests in CI.

C# .NET 10 ASP.NET Core Clean Architecture MediatR CQRS Polly FluentValidation Entity Framework Core AutoMapper

LendStream · [nanolend-app.vercel.app](#)

Micro-lending reference platform. Java 21 Spring Boot and Clojure services on Kafka with CQRS, event sourcing and a transactional outbox. React front end.

Java 21 Spring Boot Clojure React TypeScript Kafka QStash PostgreSQL Redis Docker

UniCorns, multi-tenant SaaS · [unicorns.run](#)

Multi-tenant SaaS for clinics, pharmacies and retail in Kenya. Eight Rust services on AWS Lambda, DynamoDB tenant isolation keyed on Cognito claims, offline-first web, Android and iOS clients, and four payment rails: M-Pesa, Airtel Money, Stripe and SHA insurance claims over FHIR. 2,450 automated tests.

Rust AWS Lambda DynamoDB Next.js Kotlin Claude AI M-Pesa

Vunachain, agricultural traceability

Supply chain compliance and traceability for tea and coffee exports (EU deforestation regulation, Kenya Tea Act). Django and PostGIS for geospatial checks, an event pipeline that anchors harvest records on Celo, Merkle-batched allocation of export contracts to farmers, M-Pesa payouts. Demo stage.

Rust AWS Lambda ARM64/Graviton EventBridge SQS DLQs S3 RDS PostgreSQL CloudWatch X-Ray

SwarmBench, multi-agent LLM study · developer.ericgitangu.com/pdf/research-paper.pdf

Empirical study of single-agent versus multi-agent LLM performance on software tasks, written as a Harbor benchmark. Six iterations, two verifier types, codebases up to 303K lines. Result: no multi-agent advantage for tool-using agents, reported as found.

Kimi K2.5 Fireworks AI Harbor Docker Python pandoc WeasyPrint LLM-as-judge Multi-Agent Systems Benchmark Design

AgentPlane, AI project planning · agent-plane.ericgitangu.com

Project management tool where a three-agent LangGraph pipeline breaks an epic into estimated sub-tasks with a replayable audit log. FastAPI and Next.js with cost-bounded LLM calls.

Python 3.12 FastAPI LangGraph LangChain Anthropic Next.js 14 TypeScript TanStack Query CockroachDB MongoDB Atlas

CORE TECHNICAL SKILLS

Architecture & Systems

Distributed Systems · Microservices · Event-Driven · CQRS · Event Sourcing · Resiliency & Fault Tolerance · Hexagonal / Ports & Adapters · DDD · Multi-tenant · API Design

Payments, Security & Compliance

PCI DSS (SAQ-A) · Tokenization / hosted payment sessions · Idempotency & fail-closed money paths · Settlement & reconciliation · OAuth2/JWT · OWASP Top 10 · Secure SDLC / threat modelling · MFA (WebAuthn / TOTP) · Audit hash-chains · NIST SSDF

Backend & APIs

TypeScript · Node.js · Java · Spring Boot · PHP · Python · Rust · Go · C#/ .NET · Kafka

AI/ML

TensorFlow · PyTorch · LLMs · OpenAI · Anthropic Claude · Google Gemini · RAG Systems · CNNs · GNNs · Transformers

Databases & Data

PostgreSQL · MongoDB · Redis · DynamoDB · MySQL · Kafka · RabbitMQ · Redshift · ETL/ELT Pipelines

Cloud & DevOps

AWS · GCP · Azure · Kubernetes · Docker · Terraform · CI/CD · Atomic releases / rollback · Prometheus / Grafana / Loki · OpenTelemetry

Frontend & Mobile

React · Next.js · TypeScript · Angular · Vue.js · Tailwind CSS · PWA · Kotlin + Jetpack Compose · Swift/SwiftUI · Electron

EDUCATION

Bachelor of Science in Computer Science, minor in Mathematics, University of Massachusetts at Lowell

Sep 2011 – May 2015

- Microsoft Scholar and Google Scholar. Dean's List for nine semesters. Omicron Delta Kappa.

CERTIFICATIONS & OPEN SOURCE

- **Open source and certifications:** Public GitHub repositories in Rust, Go, TypeScript, Python, Java and C#: payment and lending platforms, distributed systems, ML pipelines, developer tools. 80 certifications in AWS, Azure, GCP, AI, blockchain and application security.
- **Mentorship and teaching:** Founded a mentoring programme in 2019: 15 engineers mentored, 6 promoted to senior or lead roles. Volunteer instructor for a CodePath.org advanced algorithms cohort in 2024.